Historical Accuracy of Rolling Focus IPCA Forecasts

Abstract

- Main question: how do Focus-based IPCA forecast errors vary across rolling 3-, 6-, and 12-month horizons?
- Secondary questions:
  - How is 12-month forecaster disagreement related to error?
  - What is the performance of these forecast errors compared with an order one autoregressive rule?
- Main finding: cumulative mean absolute error (MAE) is 0.434, 0.953, and 1.794 percentage points, respectively, across 279 matched forecast months.
- State the main interpretation and its horizon-scale limitation.

1 Introduction

- Motivate the comparison of past Focus expectations with realized Brazilian IPCA.
- State the two research questions and explain the rolling forecast windows.

2 Methods

2.1 Data

This works is based mainly on two data sources:

- The expectation for inflation report from the Brazilian Central Bank, wich collects weekly data from market participants.
- The IBGE monthly realized inflation report.

2.2 Forecasts and error measures

- Compound the next 3, 6, or 12 monthly Focus medians; calculate realized inflation from matching IBGE index ratios.
- Define signed error as forecast minus realized inflation: negative means underestimation and positive means overestimation. Report both direction counts and absolute-error measures, including MAE, quantiles, and ECDFs.
- Compare the constructed 12-month forecast with the BCB's published 12-month median; use published dispersion only with the published forecast's error.

2.3 Annual target check: direction and closeness

- Treat the 25 calendar-year forecasts in the annual notebook as a separate sample from the rolling monthly panel.
- Define relative error as $|\hat{\pi}_y - \pi_y|/|\pi_y|$, where $\hat{\pi}_y$ is forecast IPCA and π_y is realized IPCA. Call a forecast "90% close" when the error is at most 0.10 (within 90–110% of the realized value), and "95% close" when it is at most 0.05. This treats under- and overestimation symmetrically.
- Show the signed-error distribution around zero and an ECDF of relative error with markers at 5% and 10%.

2.4 Simulation

- Use 10,000 paired circular moving-block bootstrap draws with 12-month blocks and seed `20261102`.
- Report percentile 95% intervals, convergence at 1,000/5,000/10,000 draws, and sensitivity to 6- and 24-month blocks.

3 Results

3.1 Main finding and descriptive statistics

Constructed horizon	MAE (p.p.)	Mean signed error (p.p.)
3 months	0.434	-0.165
6 months	0.953	-0.436
12 months	1.794	-1.033

- The 12-minus-3-month MAE difference is 1.360 p.p.; its 12-month-block bootstrap 95% interval is 0.788–2.089 p.p.
- Present the ECDFs, error distribution shapes, under/overestimation frequencies, and the other paired horizon contrasts.

3.2 Published 12-month benchmark and disagreement

- Published 12-month MAE: 1.809 p.p.; mean absolute gap from the constructed 12-month forecast: 0.049 p.p.
- Spearman correlation between published forecast dispersion and subsequent absolute error: $\rho = 0.245$ (bootstrap 95% interval: 0.046–0.413).
- Show the forecast comparison, disagreement scatter plot, bootstrap distribution, and convergence diagnostics.

3.3 Annual target check: descriptive results

- Of 25 annual forecasts, 16 underestimated realized IPCA and 9 overestimated it; mean signed error was -0.984 p.p.
- Seven forecasts (28%) were within 10% of realized inflation; one (4%) was within 5%. Show these counts on the relative-error ECDF.

4 Discussion

- Interpret the results while accounting for the different scales of cumulative 3-, 6-, and 12-month inflation.
- Keep the annual 25-year check distinct from the 279-month rolling analysis. The 90% and 95% closeness thresholds are descriptive tolerances, not confidence levels.
- Discuss overlapping windows, possible macroeconomic regime changes, aggregate Focus statistics, and historical data revisions.
- Identify questions for further analysis.

References

- Banco Central do Brasil: Focus expectations data and 12-month forecast methodology.
- Instituto Brasileiro de Geografia e Estatistica: SIDRA table 1737, annual IPCA table, and IPCA documentation.
- Add the methodological reference used for the moving-block bootstrap.

A Additional Methods and AI Use

- Add any detailed data checks and simulation diagnostics needed beyond the main article.
- Explain how AI was used in research, coding, and writing, and how the results were checked.

B Implementation

- Include key R code or point to the supplementary analysis notebooks.
- Record the Focus snapshot, IBGE source, filters, package versions, and simulation seeds.